

What is the role of advanced lipoprotein analysis in practice?



Some practitioners use advanced lipoprotein analysis with the goal of better predicting risk and individualizing lifestyle and drug therapy for cardiovascular prevention. Unfortunately, low-density lipoprotein (LDL) and high-density lipoprotein (HDL) particle number and size, other lipoprotein subfractionation, apolipoproteins B and A, and lipoprotein(a) have not yet met current standards for biomarker evaluation, and it remains to be determined whether these tests incrementally add to cardiovascular risk predicted by traditional risk factors. More importantly, it has yet to be determined whether treatment strategies guided by, or targeting, these measures improve cardiovascular outcomes. Drug therapies known to alter advanced lipoprotein analysis parameters, specifically niacin and fenofibrate, have not been shown to additionally reduce cardiovascular risk in recent randomized trials of high-risk patients treated with statin therapy. These findings suggest advanced lipoprotein analysis-guided strategies may not further reduce cardiovascular events and could lead to increased adverse effects and costs; this approach needs further research to establish its role in individualizing therapies for cardiovascular prevention. In contrast, a large body of evidence supports focusing on LDL cholesterol reduction and intensification of statin therapy to reduce cardiovascular risk. Copyright © 2012 American College of Cardiology Foundation. Published by Elsevier Inc. All rights reserved.